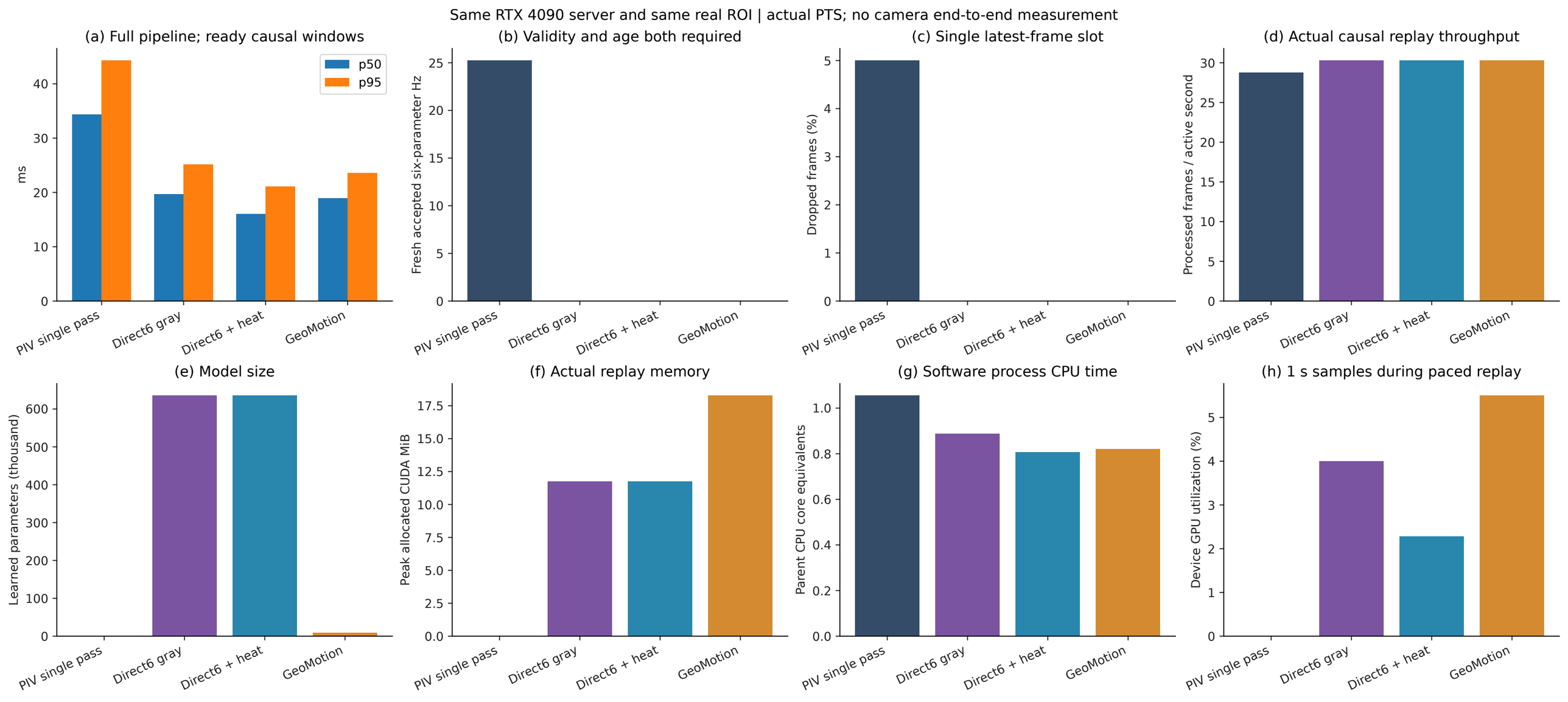
Same RTX 4090 server and same real ROI | actual PTS; no camera end-to-end measurement
(a) Full pipeline; ready causal windows
ms
p50
p95
(b) Validity and age both required
Fresh accepted six-parameter Hz
(c) Single latest-frame slot
Dropped frames (%)
(d) Actual causal replay throughput
Processed frames / active second
(e) Model size
Learned parameters (thousand)
(f) Actual replay memory
Peak allocated CUDA MiB
(g) Software process CPU time
Parent CPU core equivalents
(h) 1 s samples during paced replay
Device GPU utilization (%)
PIV single pass
Direct6 gray
Direct6 + heat
GeoMotion